Cosine and sine series of piecewise functions

Before we move on from Fourier cosine series and Fourier sine series, we want to make sure we know how to find these series representations for piecewise defined functions.

If our goal is to find the Fourier cosine series of a piecewise function $f(x)$, then we'll be finding the cosine series of its even extension, $g(x)$.

On the other hand, if our goal is to find the Fourier sine series of a piecewise function $f(x)$, then we'll be finding the sine series of its odd extension, $g(x)$.

Because the extensions are periodic, we can find their Fourier series, and because the extensions are equivalent to the original functions on $0 \leq x \leq L$, we know that the Fourier cosine series representation or Fourier sine series representation will converge to $f(x)$ on $0 \leq x \leq L$.

Let's do an example where we find the Fourier cosine series of a piecewise defined function.

Example

Find the Fourier cosine series of the piecewise function on $0 \leq x \leq L$.

$$f(x) = \begin{cases} L - x & 0 \leq x \leq \frac{L}{2} \\ L & \frac{L}{2} < x \leq L \end{cases}$$

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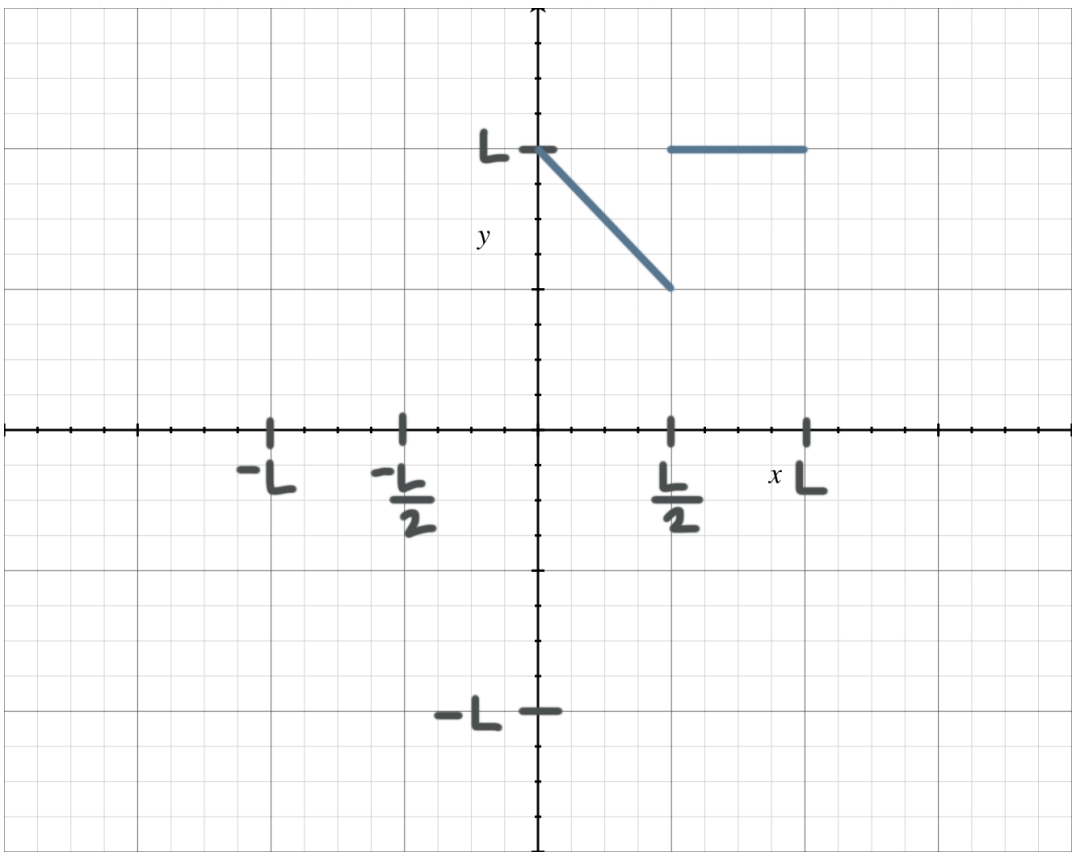


We want to build the Fourier cosine series, so we'll start by finding the even extension of the piecewise function. Because the original piecewise function $f(x)$ is defined in two pieces, its even extension $g(x)$ will be defined in four pieces.

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ f(-x) & -L \leq x < 0 \end{cases}$$

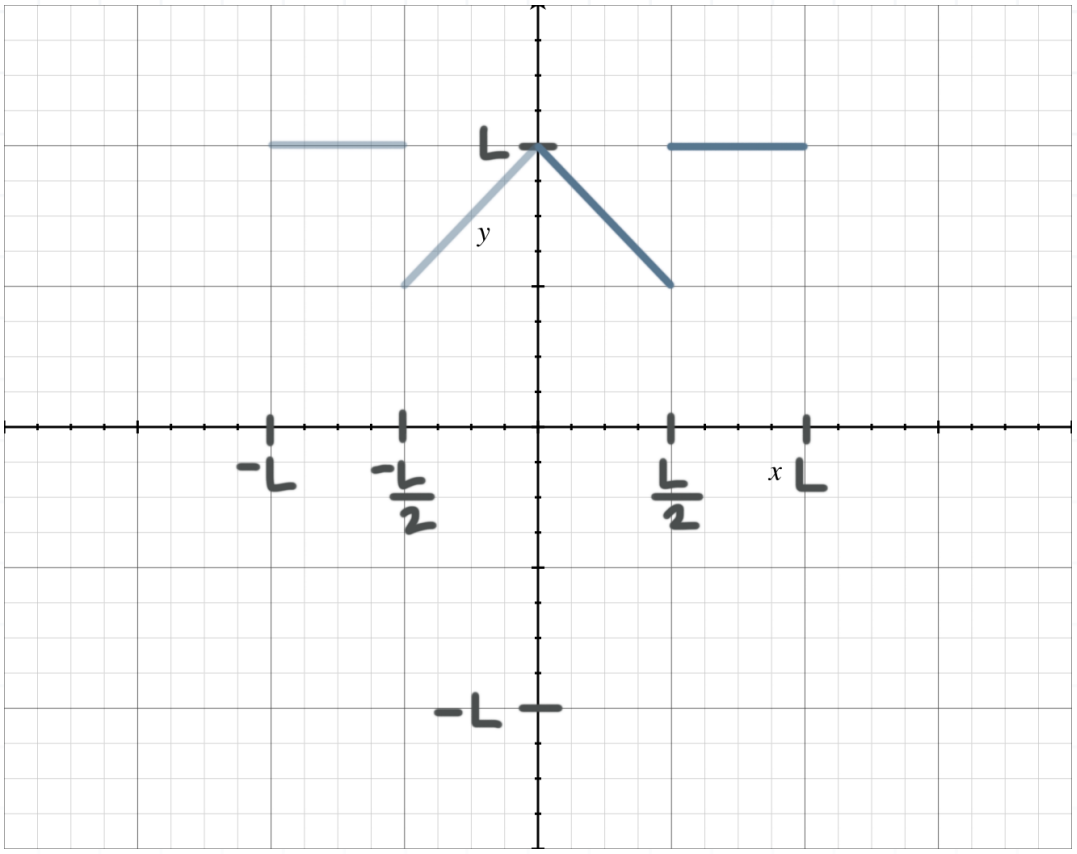
$$g(x) = \begin{cases} L & \frac{L}{2} < x \leq L \\ L - x & 0 \leq x \leq \frac{L}{2} \\ L + x & -\frac{L}{2} \leq x < 0 \\ L & -L \leq x < -\frac{L}{2} \end{cases}$$

If the graph of the original piecewise function $f(x)$ is



then the graph of its piecewise even extension $g(x)$ is

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To find the Fourier cosine series, we'll calculate A_0 and A_n. For A_0 we get

$$A_0 = \frac{1}{L}\int_0^L f(x)\,dx$$

$$A_0 = \frac{1}{L}\int_0^{\frac{L}{2}} L - x\,dx + \frac{1}{L}\int_{\frac{L}{2}}^{L} L\,dx$$

$$A_0 = \frac{1}{L}\left(Lx - \frac{1}{2}x^2\right)\Bigg|_0^{\frac{L}{2}} + \frac{1}{L}(Lx)\Bigg|_{\frac{L}{2}}^{L}$$

$$A_0 = x - \frac{1}{2L}x^2\Bigg|_0^{\frac{L}{2}} + x\Bigg|_{\frac{L}{2}}^{L}$$

$$A_0 = \frac{L}{2} - \frac{1}{2L}\left(\frac{L}{2}\right)^2 - \left(0 - \frac{1}{2L}(0)^2\right) + L - \frac{L}{2}$$

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$$A_0 = \frac{L}{2} - \frac{L}{8} + L - \frac{L}{2}$$

$$A_0 = \frac{7L}{8}$$

And for A_n we get

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{2}{L}\int_0^{\frac{L}{2}} (L-x)\cos\left(\frac{n\pi x}{L}\right)\,dx + \frac{2}{L}\int_{\frac{L}{2}}^{L} L\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = 2\int_0^{\frac{L}{2}} \cos\left(\frac{n\pi x}{L}\right)\,dx - \frac{2}{L}\int_0^{\frac{L}{2}} x\cos\left(\frac{n\pi x}{L}\right)\,dx + 2\int_{\frac{L}{2}}^{L} \cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{2L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_0^{\frac{L}{2}} - \frac{2}{L}\int_0^{\frac{L}{2}} x\cos\left(\frac{n\pi x}{L}\right)\,dx + \frac{2L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{\frac{L}{2}}^{L}$$

Use integration by parts with $u = x$, $du = dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{2L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_0^{\frac{L}{2}} - \frac{2}{L}\left[\frac{Lx}{n\pi}\sin\left(\frac{n\pi x}{L}\right) - \int \frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\,dx\right]\Big|_0^{\frac{L}{2}} + \frac{2L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{\frac{L}{2}}^{L}$$

$$A_n = \frac{2L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_0^{\frac{L}{2}} - 2\left[\frac{x}{n\pi}\sin\left(\frac{n\pi x}{L}\right) + \frac{L}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)\right]\Big|_0^{\frac{L}{2}} + \frac{2L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{\frac{L}{2}}^{L}$$

$$A_n = \frac{2L}{n\pi}\sin\left(\frac{n\pi}{2}\right) - \frac{L}{n\pi}\sin\left(\frac{n\pi}{2}\right) - \frac{2L}{(n\pi)^2}\cos\left(\frac{n\pi}{2}\right) + \frac{2L}{(n\pi)^2} + \frac{2L}{n\pi}\sin(n\pi) - \frac{2L}{n\pi}\sin\left(\frac{n\pi}{2}\right)$$

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$$A_n = \frac{2L}{n\pi}\sin(n\pi) - \frac{L}{n\pi}\sin\left(\frac{n\pi}{2}\right) - \frac{2L}{(n\pi)^2}\cos\left(\frac{n\pi}{2}\right) + \frac{2L}{(n\pi)^2}$$

For $n = 1, 2, 3, \ldots$, $\sin(n\pi) = 0$, so the expression for A_n simplifies to

$$A_n = \frac{2L}{(n\pi)^2} - \frac{L}{n\pi}\sin\left(\frac{n\pi}{2}\right) - \frac{2L}{(n\pi)^2}\cos\left(\frac{n\pi}{2}\right)$$

Then the Fourier cosine series of the piecewise function on $0 \le x \le L$ is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{7L}{8} + \sum_{n=1}^{\infty}\left[\frac{2L}{(n\pi)^2} - \frac{L}{n\pi}\sin\left(\frac{n\pi}{2}\right) - \frac{2L}{(n\pi)^2}\cos\left(\frac{n\pi}{2}\right)\right]\cos\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{7L}{8} + \frac{L}{\pi^2}\sum_{n=1}^{\infty}\left[\frac{2}{n^2} - \frac{\pi}{n}\sin\left(\frac{n\pi}{2}\right) - \frac{2}{n^2}\cos\left(\frac{n\pi}{2}\right)\right]\cos\left(\frac{n\pi x}{L}\right)$$

And now let's do one last example where we find the Fourier sine series representation of a piecewise defined function.

Example

Find the Fourier sine series of the piecewise function on $0 \le x \le L$.

$$f(x) = \begin{cases} L & 0 \le x \le \frac{L}{2} \\ x & \frac{L}{2} < x \le L \end{cases}$$

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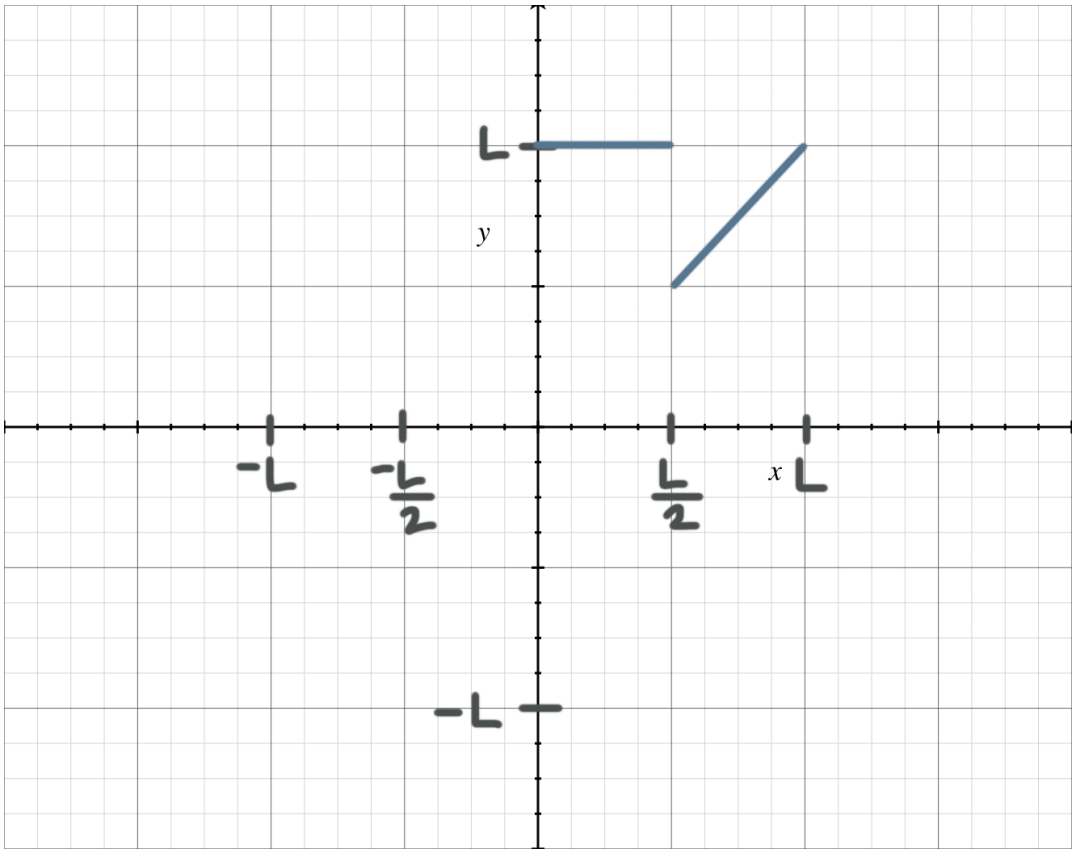


We want to build the Fourier sine series, so we'll start by finding the odd extension of the piecewise function. Because the original piecewise function $f(x)$ is defined in two pieces, its odd extension $g(x)$ will be defined in four pieces.

$$g(x) = \begin{cases} f(x) & 0 \le x \le L \\ -f(-x) & -L \le x < 0 \end{cases}$$

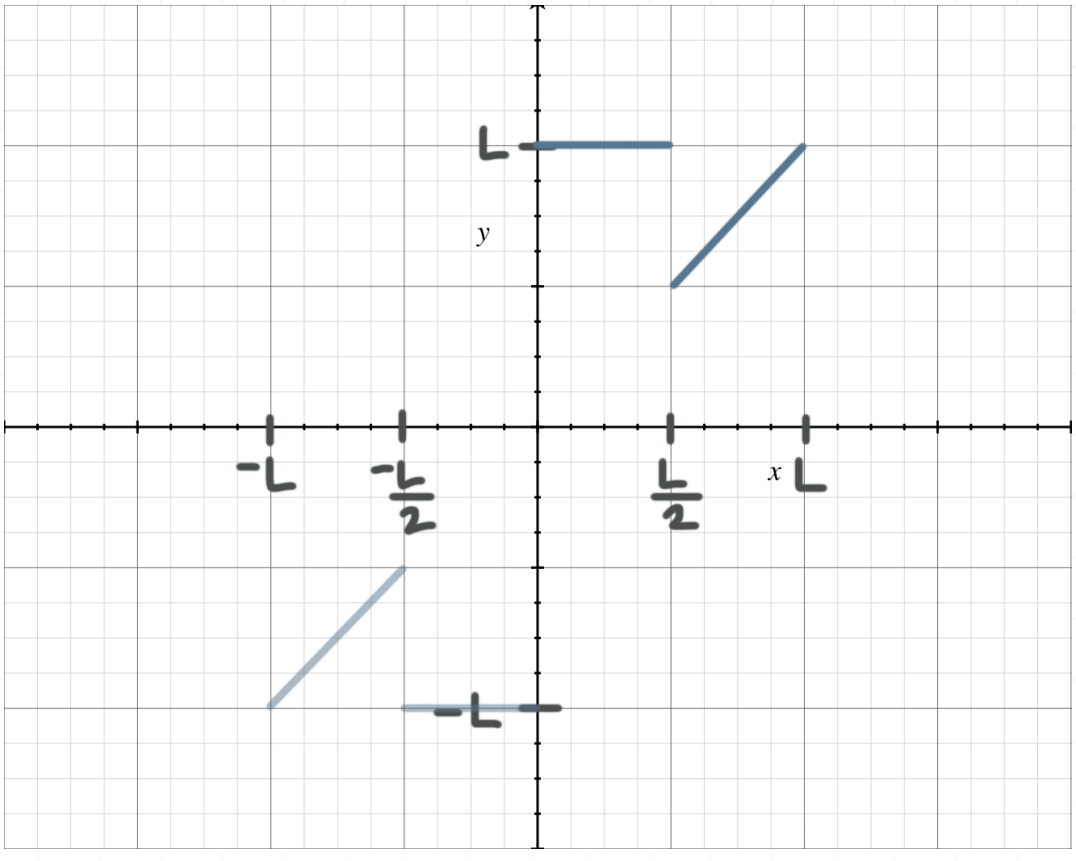
$$g(x) = \begin{cases} x & \frac{L}{2} < x \le L \\ L & 0 \le x \le \frac{L}{2} \\ -L & -\frac{L}{2} \le x < 0 \\ x & -L \le x < -\frac{L}{2} \end{cases}$$

If the graph of the original piecewise function $f(x)$ is



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then the graph of its piecewise odd extension $g(x)$ is



To find the Fourier sine series, we'll calculate B_n.

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^{\frac{L}{2}} L\sin\left(\frac{n\pi x}{L}\right)\,dx + \frac{2}{L}\int_{\frac{L}{2}}^L x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = -\frac{2}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^{\frac{L}{2}} + \frac{2}{L}\int_{\frac{L}{2}}^L x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = -\frac{2L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^{\frac{L}{2}} + \frac{2}{L}\left[-\frac{Lx}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{L}{n\pi}\int\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Big|_{\frac{L}{2}}^L$$

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$$B_n = -\frac{2}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^{\frac{L}{2}} + \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right) - \frac{2x}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_{\frac{L}{2}}^{L}$$

$$B_n = -\frac{2L}{n\pi}\cos\left(\frac{n\pi}{2}\right) + \frac{2L}{n\pi}\cos(0)$$

$$+\frac{2L}{(n\pi)^2}\sin(n\pi) - \frac{2L}{n\pi}\cos(n\pi) - \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi}{2}\right) + \frac{2L}{2n\pi}\cos\left(\frac{n\pi}{2}\right)$$

For $n = 1, 2, 3, \ldots$, $\sin(n\pi) = 0$ and $\cos(n\pi) = (-1)^n$, so the expression for B_n simplifies to

$$B_n = \frac{2L}{n\pi} + \frac{2L}{n\pi}(-1)^{n+1} - \frac{2L}{2n\pi}\cos\left(\frac{n\pi}{2}\right) - \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi}{2}\right)$$

$$B_n = \frac{2L}{n\pi}\left[1 + (-1)^{n+1} - \frac{1}{2}\cos\left(\frac{n\pi}{2}\right) - \frac{1}{n\pi}\sin\left(\frac{n\pi}{2}\right)\right]$$

Then the Fourier sine series of the piecewise function on $0 \le x \le L$ is

$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\left[1 + (-1)^{n+1} - \frac{1}{2}\cos\left(\frac{n\pi}{2}\right) - \frac{1}{n\pi}\sin\left(\frac{n\pi}{2}\right)\right]\sin\left(\frac{n\pi x}{L}\right)$$

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